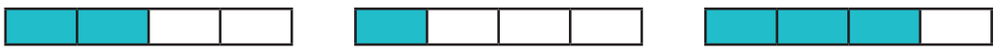
Solution

Ravi's way	Arun's way
Common Multiple of 2 and 4 = 4. Equivalent Fractions of $\frac{1}{2}=\frac{1\times2}{2\times2}=\frac{2}{4}$ Now, add $\frac{1}{2}+\frac{1}{4}=\frac{2}{4}+\frac{1}{4}$ $=\frac{2+1}{4+4}=\frac{3}{8}$ Therefore Venkat has bought $\frac{3}{8}$ litres of milk.	Common Multiple of 2 and 4 = 4. Equivalent Fractions of $\frac{1}{2}=\frac{1\times2}{2\times2}=\frac{2}{4}$ Now, add $\frac{1}{2}+\frac{1}{4}=\frac{2}{4}+\frac{1}{4}$ $=\frac{2+1}{4}=\frac{3}{4}$ Therefore Venkat has bought $\frac{3}{4}$ litres of milk.

The teacher concludes that Arun's way is correct. This can be verified by the following diagram.



$\frac{2}{4}$ + $\frac{1}{4}$ = $\frac{3}{4}$

In the above illustration note that while adding two like fractions the total number of parts (denominator) remains the same and the two shaded parts (numerator) are added.

Example 6 Add $\frac{2}{3}$ and $\frac{3}{5}$.

Solution These are unlike fractions, aren't they ? So first we need to convert them into like fractions ? Is it possible ? Yes, always. How do we do so ? The common multiple of 3 and 5 is 15. Hence, we find the equivalent fractions of $\frac{2}{3}$ and $\frac{3}{5}$ with denominator 15.

$$\frac{2}{3}=\frac{2\times5}{3\times5}=\frac{10}{15} \qquad \frac{3}{5}=\frac{3\times3}{5\times3}=\frac{9}{15}$$

$$\frac{2}{3}+\frac{3}{5}=\frac{10}{15}+\frac{9}{15}=\frac{19}{15}$$

In the above example, the common denominator is 15 (3×5). Now we observe that the numerator and denominator of the first fraction is multiplied by 5 which is the denominator of the second fraction. In the same way, the second fraction is multiplied by 3 which is the denominator of the first fraction. Now in finding the numerator of both the like fractions, we need to multiply the numerator of the first fraction by 5 and the numerator of the second by 3. In the denominator, 3×5 and 5×3 are of course the same. Thus, the technique of finding the like fraction is called ***Cross Multiplication*** technique.

That is $\frac{2}{3}\times\frac{3}{5}=\frac{(2\times5)+(3\times3)}{3\times5}=\frac{10+9}{15}=\frac{19}{15}$

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